Selection on Observables II: Estimation Approaches

PS200C, Spring 2025

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Outline

1 Matching
- Matching in X
- Measuring Distance
- Balance
- Variance Estimation
- Matching Functions
- Example: Blattman and Annan (2009)

SOO: Identification and Estimation

Recall the SOO identification assumption:

Assumption: Conditional Ignorability

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

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Today we will talk about matching, which is nearly as literal.

Matching is Not an Identification Strategy

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$$\widehat{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left(Y_i - Y_{j(i)} \right)$$

where $Y_{j(i)}$ is the outcome of an untreated observation such that $X_{j(i)}$ is the closest value to X_i among the untreated observations.

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We can also use the average of M closest matches:

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Does SOO assumption guarantee that this gets you the ATT?

Matching: Example with a Single X

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$\widehat{\tau}_{ATT} = 1/3 \cdot (6 - 9) + 1/3 \cdot (1 - 0) + 1/3 \cdot (0 - 9) = -3.7$

Common Distance Metrics

What do we mean by "closeness" in X when it is multi-dimensional?

Covariate vectors for i, j: $X_i = [X_i^{(1)}, X_i^{(2)}, ..., X_i^{(k)}]^\top$ and $X_j = [X_j^{(1)}, X_j^{(2)}, ..., X_j^{(k)}]^\top$

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Some common options:

1. Euclidean distance:

$$
\begin{aligned}
||X_i - X_j|| &= \sqrt{(X_i - X_j)^\top (X_i - X_j)} \\
&= [(X_i^{(1)} - X_j^{(1)})^2 + ... + (X_i^{(P)} - X_j^{(P)})^2]^{\frac{1}{2}}
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2. "StataD" (`nnmatch`)/default in `Match()` for `R`: rescaled Euclidean

$$StataD(X_i, X_j) = \sqrt{(X_i - X_j)^\top diag(\Sigma_X^{-1})(X_i - X_j)}$$

where Σ is the Variance-Covariance-Matrix. Invariant to rescaling of X

Common Distance Metrics

3. Mahalanobis distance:

$$MD(X_i, X_j) = \sqrt{(X_i - X_j)^\top \Sigma^{-1} (X_i - X_j)}$$

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4. GeneticD (`GenMatch`):

$$GeneticD(X_i, X_j) = \sqrt{(X_i - X_j)^\top (S^{-1/2})^\top \, W \, S^{-1/2} (X_i - X_j)}$$

where W is a $(P \times P)$ positive definite weight matrix with zeros in off-diagonals, controlling "variable importance"

Mahalanobis Distance: Example

	X_1	X_2
Treated	0	0
Control A	2	2
Control B	1.8	0

$$X_T = \begin{pmatrix} 0 & 0 \end{pmatrix}^\top \quad X_A = \begin{pmatrix} 2 & 2 \end{pmatrix}^\top \quad X_B = \begin{pmatrix} 1.8 & 0 \end{pmatrix}^\top$$

$$\Sigma = \begin{pmatrix} & X^{(1)} & X^{(2)} \\ X^{(1)} & 1 & .9 \\ X^{(2)} & .9 & 1 \end{pmatrix}$$

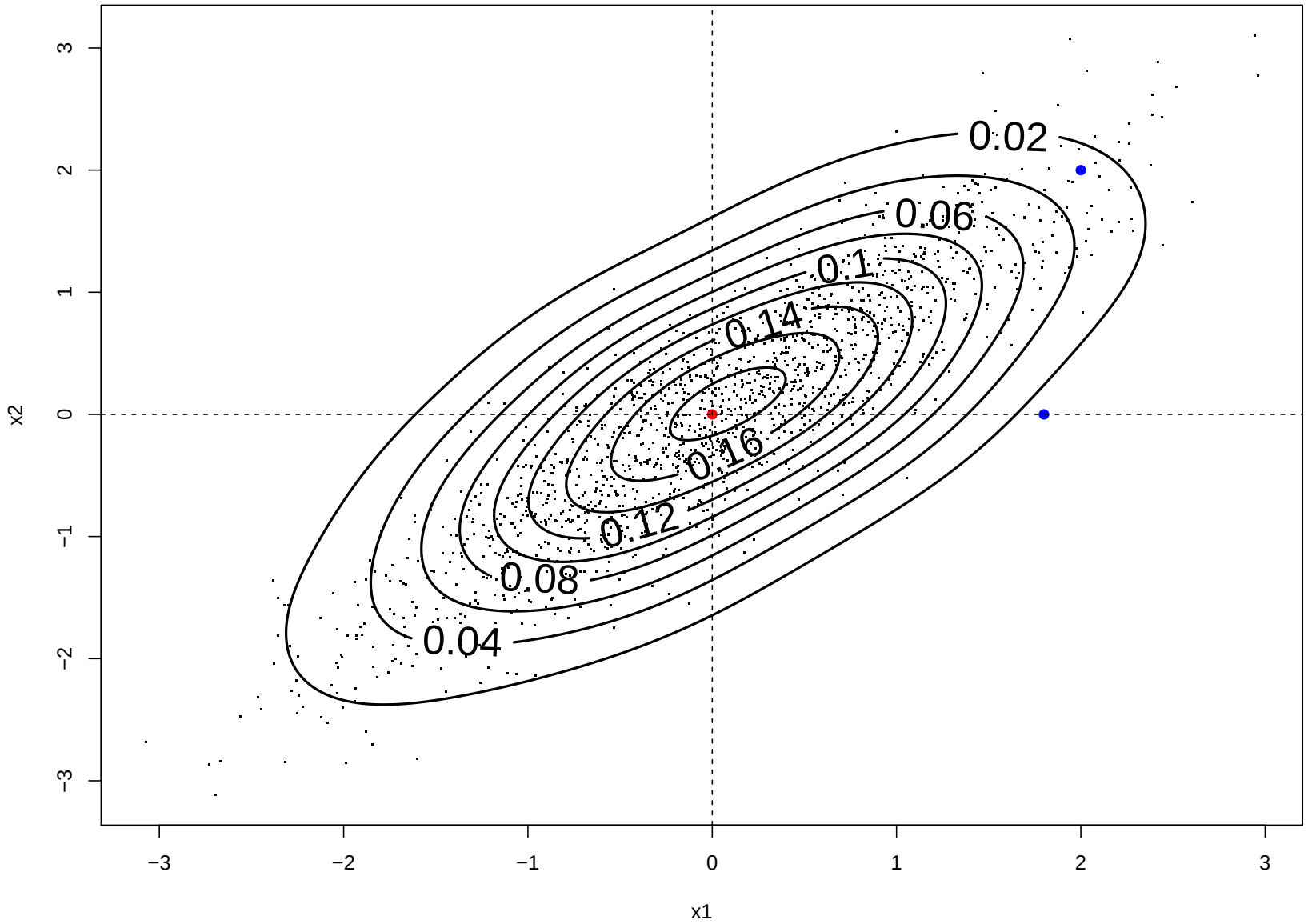
Which control is closer?

Mahalanobis Distance

$$X_T = \begin{pmatrix} 0 & 0 \end{pmatrix}^\top, \; X_A = \begin{pmatrix} 2 & 2 \end{pmatrix}^\top, \; X_B = \begin{pmatrix} 1.8 & 0 \end{pmatrix}^\top, \; \Sigma = \begin{array}{c} \\ X^{(1)} \\ X^{(2)} \end{array} \begin{array}{c} \begin{array}{cc} X^{(1)} & X^{(2)} \end{array} \\ \begin{pmatrix} 1 & .9 \\ .9 & 1 \end{pmatrix} \end{array}$$

$$\begin{aligned}
MD(X_A, X_T) &= \sqrt{(X_A - X_T)^\top \Sigma^{-1} (X_A - X_T)} \\
&= \sqrt{[\begin{pmatrix} 2 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \end{pmatrix}] \begin{pmatrix} 1 & .9 \\ .9 & 1 \end{pmatrix}^{-1} [\begin{pmatrix} 2 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \end{pmatrix}]^\top} \\
&= \sqrt{\begin{bmatrix} 2 & 2 \end{bmatrix} \begin{pmatrix} 5.2 & -4.7 \\ -4.7 & 5.2 \end{pmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix}^\top} \\
&= 4.2 \\
MD(X_B, X_T) &= \sqrt{\begin{bmatrix} 1.8 & 0 \end{bmatrix} \begin{pmatrix} 5.2 & -4.7 \\ -4.7 & 5.2 \end{pmatrix} \begin{bmatrix} 1.8 & 0 \end{bmatrix}^\top} \\
&= 17
\end{aligned}$$

Mahalanobis Distance



Local Methods and the Curse of Dimensionality

Problem: the volume increases exponentially in dim(X).

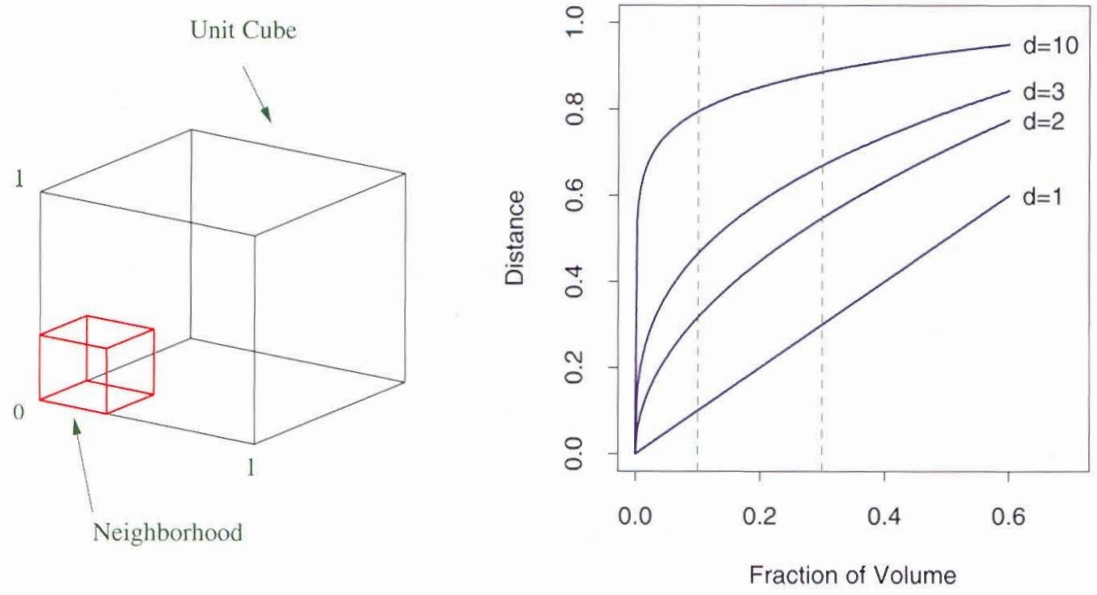


FIGURE 2.6. *The curse of dimensionality is well illustrated by a subcubical neighborhood for uniform data in a unit cube. The figure on the right shows the side-length of the subcube needed to capture a fraction r of the volume of the data, for different dimensions p. In ten dimensions we need to cover 80% of the range of each coordinate to capture 10% of the data.*

Matching with Bias Correction

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Compromise: adjust for discrepancy:

- On average, Y_{0j} differs from Y_{0i} by $\mathbb{E}[Y_0|X_i] - \mathbb{E}[Y_0|X_{j(i)}]$
- Consider population regression function $\mu_0(X) = \mathbb{E}[Y_0|X]$
- Let's estimate $\hat{\mu}_0$ by OLS regression of Y on X among controls

$$\hat{\mu}_0(X) = \beta_0 + \beta_1 X$$

Matching with Bias Correction

So subtract off estimated discrepancy on Y_0 from each pair (Abadie & Imbens, 2005)

$$\tilde{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} ((Y_i - Y_{j(i)}) - (\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_{j(i)})),$$

- these "bias-corrected" matching estimators are an improvement even if $\hat{\mu_0}$ is misspecified
- the large sample distribution of this estimator (for the case of matching with replacement) is roughly normal.

In R: `Match(Y,Tr, X,BiasAdjust = TRUE)`

Bias Adjustment with Matched Data

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Estimate $\widehat{\mu}_0(x) = \beta_0 + \beta_1 x = 5 - .4x$. Now plug in:

$$\begin{aligned}
\widehat{\tau}_{ATT} &= 1/3\{((6-9) - (\widehat{\mu}_0(3) - \widehat{\mu}_0(3))) \\
&+ ((1-0) - (\widehat{\mu}_0(1) - \widehat{\mu}_0(2))) \\
&+ ((0-1) - (\widehat{\mu}_0(10) - \widehat{\mu}_0(8)))\} \\
&= \frac{1}{3}[(6-9) + [(1-0) - (4.6-4.2)] + [(0-1) - (1-1.8)]] \\
&= -0.87 \ \text{(rather than -1)}
\end{aligned}$$

Checking balance: Theory

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One workflow:

- Estimate $\rightarrow$ check balance $\rightarrow$ re-estimate $\rightarrow$ check balance $\rightarrow \cdots$ (ad infinitum until you get a good balance)
- Is this data snooping? No, because blind to Y

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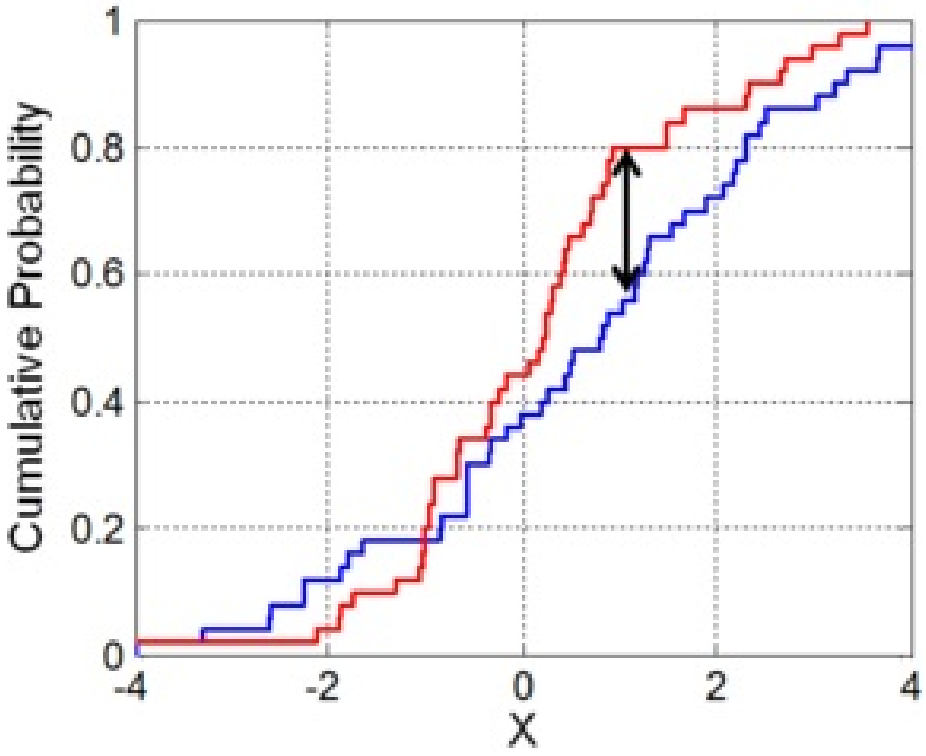

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Reject the null at level α if

$$\frac{D}{\sqrt{(n_1 + n_0)/n_1 n_0}} > c_\alpha$$

level (α)	.1	.05	.01
critical value (c_α)	1.22	1.36	1.63

Balance Checks

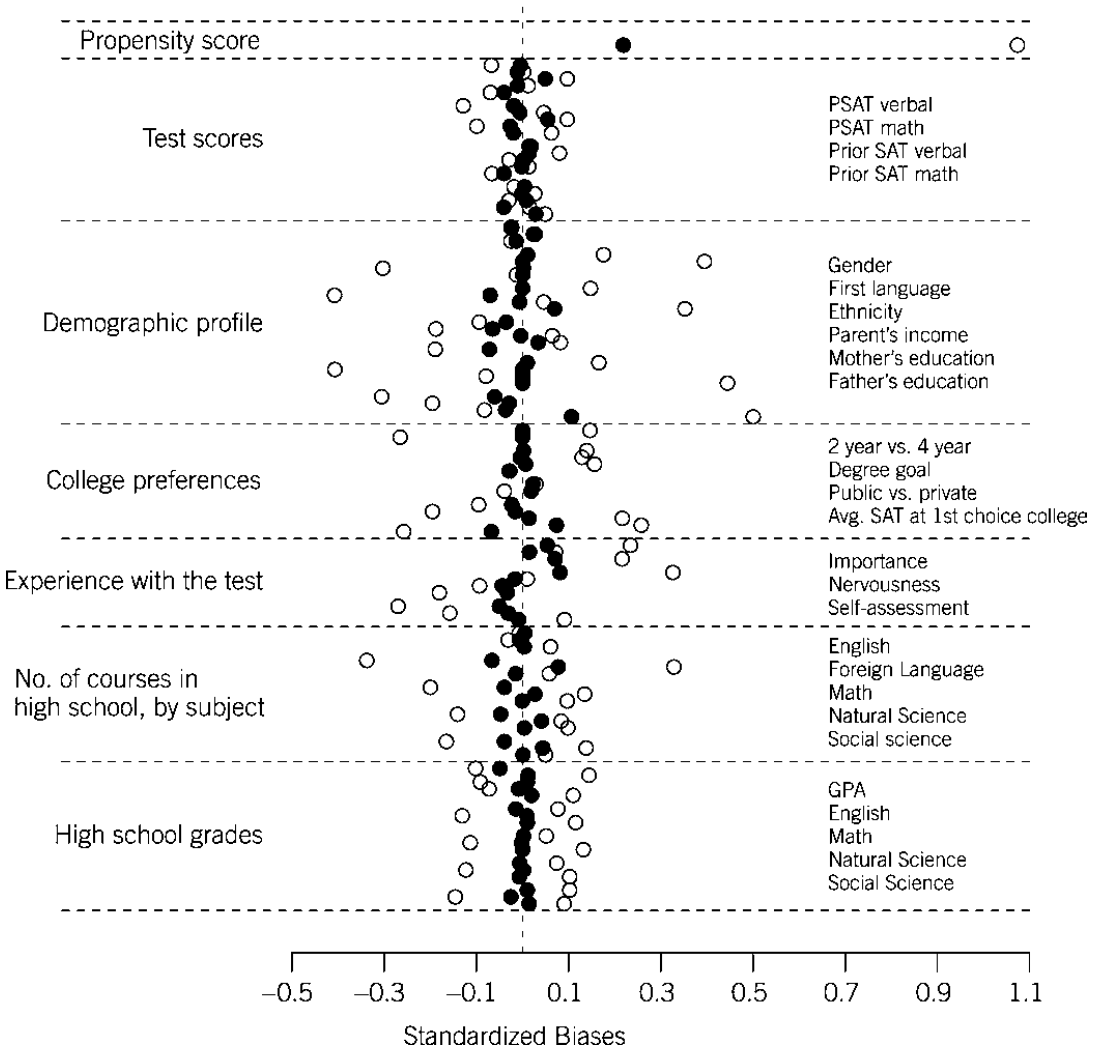

Figure 3. Standardized Biases Without Stratification or Matching, Open Circles, and Under the Optimal [.5, 2] Full Match, Shaded Circles.

Balance Checks

TABLE 2. Balance Summary Statistics and Tests: Russian and Chechen Sweeps

Pretreatment Covariates	Mean Treated	Mean Control	Mean Difference	Std. Bias	Rank Sum Test	K-S Test
Demographics						
Population	8.657	8.606	0.049	0.033	0.708	0.454
Tariqa	0.076	0.048	0.028	0.104	0.331	—
Poverty	1.917	1.931	−0.016	−0.024	0.792	1.000
Spatial						
Elevation	5.078	5.233	−0.155	−0.135	0.140	0.228
Isolation	1.007	1.070	−0.063	−0.096	0.343	0.851
Groznyy	0.131	0.138	−0.007	−0.018	0.864	—
War Dynamics						
TAC	0.241	0.282	−0.041	−0.095	0.424	—
Garrison	0.379	0.414	−0.035	−0.072	0.549	—
Rebel	0.510	0.441	0.070	0.139	0.240	—
Selection						
Presweep violence	3.083	3.117	−0.034	0.009	0.454	0.292
Large-scale theft	0.034	0.055	−0.021	−0.115	0.395	—
Killing	0.117	0.090	0.027	0.084	0.443	—
Violence Inflicted						
Total abuse	0.970	0.833	0.137	0.124	0.131	0.454
Prior sweeps	1.729	1.812	−0.090	−0.089	0.394	0.367
Other						
Month	7.428	6.986	0.442	0.130	0.260	0.292
Year	2004.159	2004.110	0.049	0.043	0.889	1.000

Note: 145 matched pairs. Matching with replacement.

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- General rule: use whatever gives you the best balance subject to conditioning on what you need and not dropping too many units.
- Adjustment methods help, but rely on a model to impute missing potential outcomes.

Variance of the Matching Estimator

Option 1. Ignore the matching uncertainty and estimate the SEs from whatever model you run on the matched sample

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Option 4. Post-treatment model SE

- Use matching to get weights, run weighted regression, cluster robust SEs
- Default in `MatchIt` package; good idea to use it.

Useful Matching Functions

Two packages widely used: `MatchIt`, and `Matching`.
We'll use `Matching` today (the `Match` function), but provide `MatchIt` examples elsewhere.

```
Match(Y = NULL, Tr, X, Z = X, V = rep(1, length(Y)),
    estimand = "ATT", M = 1, BiasAdjust = FALSE, exact = NULL,
    caliper = NULL,  replace = TRUE, ties = TRUE,
    CommonSupport = FALSE, Weight = 1, Weight.matrix = NULL,
    weights = NULL...)
```

Default distance metric is normalized Euclidean distance

- `MatchBalance` for balance checking

Example: Blattman and Annan (2009)

The Consequences of Child Soldiering. The Review of Economics and Statistics.

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ID Strategy: SOO. According to the logic by which abduction occurred, we will assume that abduction is indiscriminate, conditional on age and location.

> *Abduction was large-scale and seemingly indiscriminate; 60,000 to 80,000 youth are estimated to have been abducted and more than a quarter of males currently aged 14 to 30 in our study region were abducted for at least two weeks...*
>
> *Youth were typically taken by roving groups of 10 to 20 rebels during night raids on rural homes. Adolescent males appear to have been the most pliable, reliable and effective forced recruits, and so were disproportionately targeted by the LRA. Youth under age 11 and over 24 tended to be avoided and had a high probability of immediate release.*

Example: Blattman and Annan (2009)

Data: panel survey of male youth in war-afflicted regions of Uganda.

- `abd`: abducted by the LRA (the treatment)
- `c_ach - c_pal`: Location indicators
- `age`: age in years
- `fthr_ed`,`mthr_ed` : father's/mother's education (years)
- `orphan96`: indicator if parent's died before 1997
- `hh_fthr_frm`: indicator if father is a farmer
- `hh_size96`: household size in 1996
- `educ`: years of education
- `distress`: index of emotional distress (0-15)
- `logwage`: log of average daily wage earned in last 4 weeks

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After matching, which of these should you expect good balance on once NB: `educ`, `distress`, and `logwage` are measured after abduction.

Code example(s)

Check online for example code of matching and balance testing, using both older and newer packages.